Discrete Math Midterm Solutions

1. (a) How many permutations are there of the letters $a, b, c, d, e, f, g, h, i$, and j?

Solution

10!

(b) How many start with the letter a?

Solution

There are 9! ways to arrange the other 9 letters, so 9!

(c) How many of the permutations from (a) start with a vowel (a, e or i) ?

Solution

There are 3 ways to choose the first letter, and then 9! ways to choose the others. So there are $3 \cdot 9!$ ways in total.

(d) How many of the permutations from (a) have no adjacent vowels?

Solution

We have 7! ways to arrange the constanants, $\binom{8}{3}$ ways to choose the spaces between them in which to put the vowels, and 3! ways to put the vowels in these spots. Thus there are $7! \cdot 3! \cdot \binom{8}{3} = 7! \cdot P(8, 3)$ ways.

(e) How many of the permutations from (a) have all vowels together?

Solution

Gluing the vowels together there are 8 elements, so 8! ways to arrange them. There are 3! ways to glue the vowels together, so $8! \cdot 3!$ arrangements in total.

2. (a) Complete the definitin of isomorphism. Two graphs G and H are isomorphic, written $G \cong H$, if there exists an isomorphism f from G to H; that is, a function...

Solution

$f : V(G) \to V(H)$ such that

- f is bijective, and
- $u \sim v$ if and only if $f(u) \sim f(v)$ for any $u, v \in V(G)$.

(b) Show that $F \cong G$ and $G \cong H$ imply $F \cong H$.

Solution

Assume the premises. Then by definition, there exist isomorphisms $f : V(F) \to V(G)$ and $g : V(G) \to V(H)$. We show that $g \circ f : V(F) \to V(H)$ is an isomorphism of F to H. Indeed $g \circ f$ is bijective as g and f are.

Now let u, v be vertices of F. As f is an isomorphism, $u \sim v \iff f(u) \sim f(v)$. As g is an isomorphism, $f(u) \sim f(v) \iff g(f(u)) \sim g(f(v))$. So by logical syllogism $u \sim v \iff g(f(u)) \sim g(f(v)) \iff g \circ f(u) \sim g \circ f(v)$, as needed.

3. Let G be the shown graph.

   How many ways are there to colour the seven vertices of the unlabelled graph G with seven distinguishable colours, so that each vertex gets a different colour?

Solution

There are 7! colourings, and 12 automorphisms of the graph, so 7!/12 different colouring upto isomorphism.

4. (a) How many different r-paths are there in K_n? (Recall r-paths have $r + 1$ vertices.)

Solution

Each $r + 1$ permutation of $[n]$ designates a path, by designating the vertices and the order they are visited. But this counts each path twice, as either endpoint can be chosen first. So there are $P(n, r + 1)/2 = n!/[2 \cdot (n - r - 1)!]$ such paths.

(b) How many different r-paths are there in C_n?

Solution

n

(c) How many different 4-paths are there in Q_n ($n \geq 4$)? (Hard. Do this last.)

Solution

Vertices of Q_n are binary strings in $\{0,1\}^n$. To describe and edge, $v_i v_{i+1}$ we can designate the vertex v_i and the bit $b_{i+1} \in [n]$ that where v_i and v_{1+1} differ. So a 4-path $v_0, v_1, \ldots, v_4$ in Q_n is described by its first vertex v_0, and four bits b_1, b_2, b_3, b_4. **The vertex v_0 can be chosen in 2^n ways.**
Not every choice of bits yields a path. We do **not** get a path if

- two consecutive bits b_i and b_{i+1}, (because then the $i-1$ and $i+1$ vertices of the path are the same).
- the bits are i, i, j, j for integers i and j, (because then then $v_4 = v_0$).

Otherwise, we get a path. So either our bits are all distinct, so **can be chosen in $P(n,4)$ ways**, or they are i, i, j, k **chosen in $\binom{n}{3} \cdot 3$ ways** and must be arranged without the is adjacent, which **can be done in 3 ways**. Finally, each path has been counted twice as either endpoint could be the first one. So all together there are

$$2^n \dot{(} P(n,4) + 9\binom{n}{3})/2 = 2^{n-1}(P(n,4) + 9\binom{n}{3})$$

such paths.

5. Fill in the blanks with the appropriate arrows: $\Rightarrow$, $\Leftarrow$, or $\Longleftrightarrow$.
For a given universe U and open statements $p(x)$ and $q(x)$ over that universe:

(a) $\forall x[p(x) \vee q(x)]$ _____ $\forall x[p(x)] \vee \forall x[q(x)]$

(b) $\exists x[p(x) \wedge q(x)]$ _____ $\exists x[p(x)] \wedge \exists x[q(x)]$

(c) $\exists x[p(x) \vee q(x)]$ _____ $\exists x[p(x)] \vee \exists x[q(x)]$

(d) $\neg \forall x[p(x)]$ _____ $\exists x[\neg p(x)]$

Solution

(a) $\Leftarrow$

(b) $\Rightarrow$

(c) $\Longleftrightarrow$

(d) $\Longleftrightarrow$

6. Show whether or not the following arguments are valid.

(a)

1	$p \wedge q$
2	$p \to (r \wedge q)$
3	$r \to (s \vee t)$
4	$\neg s$
$\therefore$	t

(b)

1	$p \to (q \to r)$
2	$p \vee s$
3	$t \to q$
4	s
$\therefore$	$\neg r \to \neg t$

Solution

(a)

1	$p \wedge q$	given
2	p	conjunctive simplification
3	$p \to (r \wedge q)$	given
4	$r \wedge q$	2 and 3 and Modus Ponens
5	r	conjunctive simplification
6	$r \to (s \vee t)$	given
7	$s \vee t$	Modus Ponens
8	$\neg s$	given
$\therefore$	t	7,8 and disjunctive syllogism

(b) It is not valid. Let $t = s = q = 1$ and $r = p = 0$. Then the premises are true but the conclusion is false.

7. Prove without a Venn Diagram that

$$A \subset B \iff A \cap \bar{B} = \emptyset.$$

Solution

By definition $A \subset B$ means that for all $x, x \in A \to x \in B$. But

$$\begin{aligned}
x \in A \to x \in B \quad &\iff \neg(x \in A) \vee (x \in B) \text{ law of implication} \\
&\iff x \notin A \vee x \in B \\
&\iff x \in \bar{A} \vee x \in B \text{ def of complement} \\
&\iff x \in \bar{A} \cup B \text{ def of cup} \\
&\iff x \notin \overline{\bar{A} \cup B} \text{ def of complement} \\
&\iff x \notin A \cap \bar{B} \text{ DeMorgan and double neq}
\end{aligned}$$

So $A \subset B$ is true if and only if for all x, $x \notin A \cap \bar{B}$. That is to say if and only if $A \cap \bar{B} = \emptyset$.

8. You choose 2 Jellybellies from a cup of 30 different Jellybellies, among which are the flavours banana and lemon. What is the probability of choosing

   (a) banana and lemon

Solution

Consider a universe of $\binom{30}{2}$ unordered choices, we get a probability of $1/\binom{30}{2}$. Or considering a universe of $30 \cdot 29$ ordered choices, the event contains two outcomes, so we get a probability of $2/(30 \cdot 29) = 1/\binom{30}{2}$.

(b) banana but not lemon

Solution

$\binom{28}{1}/\binom{30}{2} = 28/\binom{30}{2} = (2 \cdot 28)/(29 \cdot 30)$.

(c) neither banana or lemon

Solution

$\binom{28}{2}/\binom{30}{2} = (28 \cdot 27)/(30 \cdot 29)$

9. Where F_i is the ith Fibonacci number, show that the following is true for all $n \geq 0$.

$$F_0 + F_1 + \cdots + F_n = F_{n+2} - 1.$$

Solution

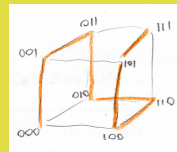
For the base step, observe that $F_0 = 1 = 2 - 1 = F_2 - 1$. Now assume that the result has been proved for all $n < k$, we prove if for $n = k$. Indeed, by the induction hypothesis $F_0 + F_1 + \cdots + F_{k-1} + F_k = F_k + F_{k+1} - 1$ and this is $F_{k+2} - 1$, by the definition of the Fibonacci numbers.

10. (a) Find a Gray code of order 3 that begins with the emptyset and ends with the full set $[3] = \{1, 2, 3\}$.

(b) Show that there cannot be a Gray code of 4 starting at the empty set and ending at $[4]$.

Solution

It's useful to view a Gray code as a path in Q_n that contains all vertices. Here is the order 3 Gray code we want



It is $(000, 001, 011, 010, 110, 100, 101, 111)$.

There cannot be such a code of order 4. In a Gray code (represented as binary strings), consecutive strings differ by only one bit. A Gray code of order 4 has 2^4 strings, so to get from the first to the last, we have to change $2^4 - 1 = 15$ bits. As this is odd, the first and last strings differ in the parity of the number of ones. But 0000 and 1111 both have an even number of $'1'$s.